

Assignment: Detecting Swimming Pools from Aerial Images using YOLO26 and RF-DETR

Introduction

Object detection from aerial imagery is a valuable computer vision task with real-world applications ranging from urban planning to commercial analytics. In this assignment, you will build a swimming pool detector using state-of-the-art object detection architectures, including YOLO26 and RF-DETR.

Your first task will be to annotate an unlabeled aerial imagery dataset using GroundingDINO. You will then train multiple object detection models using transfer learning and compare their performance across model scales and architectures.

As an advanced extension, you will implement Oriented Bounding Box (OBB) detection for improved localization of rotated swimming pools.

Rationale

Swimming pool detection can support applications such as:

Commercial Targeting

- Pool cleaning services
- Smart pool equipment
- Renovation and maintenance services

Insurance and Tax Assessment

- Property auditing
- Risk assessment

Urban Monitoring

- Recreational infrastructure tracking
 - Water usage estimation
 - Land-use analytics
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Objectives

1. Annotation Pipeline

To accelerate annotation, students must:

- Use GroundingDINO for automatic annotation
- Review and manually correct all annotations
- Export annotations in YOLO format

Students may use:

- Roboflow annotation tools (online)
- FiftyOne from Voxel51 (Python library)

Important: Roboflow may only be used for annotation assistance, not for model training.

2. YOLO26 Training and Scaling

Train YOLO26 models using transfer learning.

Available YOLO26 variants:

- YOLO26n
- YOLO26s
- YOLO26m
- YOLO26l
- YOLO26x

Requirements:

- At least **one** model size is mandatory
- Comparison across multiple sizes is strongly encouraged

Students must compare:

- Accuracy
 - Robustness
 - Parameter count
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3. RF-DETR Benchmarking

Train and evaluate RF-DETR models.

Recommended variants:

- RF-DETR Nano
- RF-DETR Small

Students must compare RF-DETR against YOLO26 models, with particular attention to:

- Small object detection
 - Generalization performance
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4. Oriented Bounding Boxes (OBB)

Implement rotated object detection using:

- YOLO26-OBB

Students must compare:

- Standard horizontal bounding boxes
- Oriented bounding boxes (OBB)

Discussion points:

- Localization precision
 - Failure cases
 - Handling of rotated pool geometries
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Materials Provided

Dataset

Aerial images containing unlabeled swimming pools:

- [Google Drive Dataset Folder](#)

Example Notebooks

- GroundingDINO annotation guide
- YOLO fine-tuning example
- RF-DETR fine-tuning example

The notebooks are attached in the same folder as the assignment.

Task Instructions

Step 1 — Annotate Images Using GroundingDINO

Students must:

1. Load the aerial image dataset
2. Run GroundingDINO using prompts such as:
 - "swimming pool"
 - "pool"
 - "outdoor swimming pool"
3. Export annotations in YOLO format

Manual Verification (MANDATORY)

Students **must**:

- Manually review all annotations
- Fix incorrect labels
- Remove false positives
- Add missed pools

Important: Automatic annotations are intentionally imperfect.

Step 2 — Train YOLO26 Models

Transfer Learning

Load pretrained YOLO26 models.

Requirements:

- At least one YOLO26 model is mandatory
 - Comparing multiple scales is recommended
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Training

Students must:

- Create train/validation splits
- Justify the split strategy
- Configure training hyperparameters

Required reported parameters:

- Optimizer
 - Learning rate
 - Number of epochs
 - Image size
 - Batch size
 - Data augmentations
 - Learning rate scheduler
 - Hardware used
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Evaluation

Students must report:

- mAP50
 - mAP50-95
 - Precision
 - Recall
 - Parameter count
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Step 3 — Train RF-DETR Models

Train:

- RF-DETR Nano
- RF-DETR Small (optional)

Students must evaluate:

- Transformer vs YOLO architectures
 - Detection robustness
 - Small pool detection performance
 - Generalization to unseen images
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Step 4 — Oriented Bounding Box Detection

Train:

- YOLO26-OBB
- YOLO11-OBB

Students must:

- Convert annotations to OBB format
 - Compare OBB vs horizontal bounding boxes
 - Analyze localization precision
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Required Experimental Analysis

Students must discuss:

1. Which architecture performed best?
 2. Which model size offered the best speed/accuracy tradeoff?
 3. Did RF-DETR outperform YOLO-based detectors?
 4. When are OBB annotations beneficial?
 5. Which failure cases occurred most often?
 6. Which augmentations improved performance the most?
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Failure Analysis Section

Students must provide:

- At least 5 failure cases
- Visual explanations of failures
- Proposed solutions for each failure case